

MRC Lecture 6 — Derivation Guide 3

The Contrastive Objective Is Unique

Sai T. Reddy — ETH Zurich / BIIE

Sources: *Methods for Molecular Recognition Computing, Supplementary Note S3; Computational Convergence, Supplementary Note S2 (thymic selection, AIRE-driven negative selection).*

1. Motivation

WB1 established that the convergence equation is the unique selection function. WB2 showed that, with a bilinear molecular score, the equation becomes the transformer attention mechanism and the dual-encoder architecture assembles itself. This guide completes the third level of inevitability: the **training objective** is also prescribed. The InfoNCE loss is not one contrastive loss among many — it is the **maximum-likelihood estimator** of the convergence equation. Train with anything else and you are fitting the wrong selection model.

We derive InfoNCE from the convergence equation, work a numerical example showing the loss falling from an untrained value of 1.10 to a trained value of 0.019, derive the symmetric form from binding reciprocity, explain why the standard alternatives fail, and close with the observation that the adaptive immune system has been running this exact algorithm — contrastive learning with hard negatives — since long before it had a name.

2. From the Convergence Equation to a Loss

Training data is a set of N observed **true binding pairs** $\{(A_i, T_i)\}_{i=1}^N$ — an agent and the target it actually binds. We assemble them into a batch and compute the $N \times N$ matrix of compatibility scores $s_{ij} = \mathbf{q}_i \cdot \mathbf{k}_j / \tau$, where $\mathbf{q}_i = \text{Encoder}_A(A_i)$ and $\mathbf{k}_j = \text{Encoder}_T(T_j)$.

For agent i , the other targets in the batch $\{T_l: l \neq i\}$ serve as the competing pool — the **in-batch negatives**. By the convergence equation, the probability that agent i selects its true partner T_i out of the batch is:

$$P_i = \frac{\exp(s_{ii})}{\sum_{l=1}^N \exp(s_{il})}$$

The true pair is the diagonal entry s_{ii} ; the denominator sums over the whole row (the true target plus all in-batch negatives).

Maximum likelihood. Assuming the pairs are independent, the likelihood of the entire observed batch is the product $\prod_i P_i$. The maximum-likelihood objective is to maximize this product, equivalently to minimize the negative log-likelihood:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log P_i = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(s_{ii})}{\sum_{l=1}^N \exp(s_{il})}$$

This is exactly the InfoNCE loss. It is not selected from a menu of contrastive losses for empirical performance — it is the negative log-likelihood of the convergence equation. Given that the convergence equation is the unique selection model (WB1), its MLE is the unique correct objective.

3. Numerical Example: Untrained vs. Trained

Take a small batch of $N = 3$ pairs. We compare an untrained model (uninformative scores) to a trained model (scores that separate true pairs from negatives).

Untrained model

Before training, the encoders produce uninformative embeddings; all scores are roughly equal. Take every $s_{ij} = 0$. Then for each agent:

$$P_i = \frac{\exp(0)}{\exp(0) + \exp(0) + \exp(0)} = \frac{1}{3}$$

The per-example loss is:

$$\mathcal{L} = -\log\left(\frac{1}{3}\right) = \log 3 = \boxed{1.10}$$

The model is at chance — it cannot distinguish the true target from the negatives. A loss of $\log N$ is the signature of an untrained contrastive model on a batch of size N .

Trained model

After training, the encoders place each agent's embedding close to its true partner and far from negatives. Suppose the diagonal (true-pair) score rises to $s_{ii} = 4.64$ while the off-diagonal (negative) scores stay near $s_{il} = 0$. Then:

$$P_i = \frac{\exp(4.64)}{\exp(4.64) + \exp(0) + \exp(0)} = \frac{103.5}{103.5 + 1 + 1} = \frac{103.5}{105.5} = 0.981$$

The per-example loss is:

$$\mathcal{L} = -\log(0.981) = \boxed{0.019}$$

The loss falls from 1.10 (chance) to 0.019 (confident correct retrieval) as the model learns to separate true pairs from in-batch negatives. The entire training process is the encoders learning embeddings that push the diagonal scores up and the off-diagonal scores down — concentrating the convergence-equation probability on the true pairs.

4. The Symmetric Loss From Binding Reciprocity

Binding is reciprocal: the free energy of the A – T complex is a single physical quantity,

$$\Delta G_{AT} = \Delta G_{TA}$$

so the compatibility is symmetric. This means the model should predict equally well in both directions: given an agent, retrieve its target; and given a target, retrieve its agent. The InfoNCE loss as written in §2 is the **agent** → **target** direction (row-wise softmax over the score matrix). The **target** → **agent** direction is the column-wise softmax:

$$P_j^{(\text{col})} = \frac{\exp(s_{jj})}{\sum_l \exp(s_{lj})}, \quad \mathcal{L}_{T \rightarrow A} = -\frac{1}{N} \sum_j \log P_j^{(\text{col})}$$

The **symmetric loss** averages both directions:

$$\mathcal{L}_{\text{sym}} = 1/2 (\mathcal{L}_{A \rightarrow T} + \mathcal{L}_{T \rightarrow A})$$

This is the same bidirectional contrastive loss used by CLIP (Radford et al., 2021) — but where CLIP adopted it empirically, here it is **derived** from the physical reciprocity of binding. The symmetric form is prescribed, not chosen.

5. Why the Alternatives Fail

The standard contrastive and pairwise losses are maximum-likelihood estimators of *different* probability models, each of which violates one of the five axioms (WB1):

- **Triplet loss** (anchor, positive, negative with a margin): compares the true pair against one negative at a time, with no normalization over the full pool. It violates **Axiom 2 (normalization)** — it does not define a proper probability distribution over the candidate set. It is the MLE of no consistent selection model.
- **Binary cross-entropy** (independent bind/no-bind per pair): treats each agent–target pair as an independent Bernoulli event. This violates **Axiom 5 (IIA)** — the predicted probability for one pair does not depend on the competing pool, so the relative odds between two targets are not

coupled through a shared partition function. It is the MLE of a model in which targets do not compete, contradicting the physics of competitive binding.

- **Margin / hinge loss:** imposes a hard separation gap. This violates the soft, stochastic nature of equilibrium selection (every candidate has nonzero probability — **Axiom 1**), replacing the graded Boltzmann distribution with a hard threshold.

None of these is “worse” merely in the sense of empirical benchmark performance. Each is the *correct* estimator of an *incorrect* selection model. InfoNCE is the correct estimator of the correct model — the convergence equation. This is why, when building an SFM, the objective is not a hyperparameter to tune: it is fixed by the physics.

6. The Immune System Was Already Doing This

The structure of contrastive training — a true pair, a pool of competing negatives, and special “hard” negatives that look deceptively like targets — is not merely analogous to adaptive immunity. It is the algorithm the thymus runs to train the T-cell repertoire. (Computational Convergence, Supplementary Note S2, develops this correspondence.)

During T-cell development in the thymus, a developing T cell bearing a particular T-cell receptor (TCR) is presented with self-peptides displayed on MHC molecules. The cell must learn to respond to foreign peptides while *not* responding to self. The key mechanism: **AIRE** (the autoimmune regulator) drives the expression of thousands of tissue-restricted self-antigens in the thymic medulla — proteins that would otherwise only appear in peripheral organs. T cells that bind these self-antigens too strongly are deleted (**negative selection**). These AIRE-driven self-antigens are precisely the **hard negatives**: peptides that resemble potential targets but must be learned as non-targets.

Contrastive learning	Thymic selection
Anchor (agent)	Developing T cell (its TCR)
Positive pair	Cognate antigen the TCR should recognize
In-batch negatives	Other self-peptides present in the thymus
Hard negatives	AIRE-driven tissue-restricted self-antigens (negative selection)
InfoNCE objective	Repertoire selection (survival vs. deletion)
Temperature τ	Selection stringency / signaling threshold
Trained encoder	The mature, self-tolerant TCR repertoire

The thymus implements contrastive learning with hard negatives to produce a self-tolerant, pathogen-reactive T-cell repertoire. AIRE’s biological function — surfacing the deceptive self-antigens that naive negative selection would miss — is exactly the function of hard-negative mining in contrastive training: forcing the model to learn the difficult distinctions, not just the easy ones. The immune system was running this algorithm for hundreds of millions of years before Radford et al. wrote down InfoNCE.

7. Summary

Concept	Result	Significance
InfoNCE derivation	$\mathcal{L} = -\frac{1}{N} \sum_i \log \frac{\exp(s_{ii})}{\sum_l \exp(s_{il})}$	MLE of the convergence equation
Untrained loss	$P = 1/3, \mathcal{L} = \log 3 = 1.10$	Chance baseline = $\log N$
Trained loss	$P = 0.981, \mathcal{L} = 0.019$	True pairs separated from negatives
Symmetric loss	$\mathcal{L}_{\text{sym}} = \frac{1}{2} (\mathcal{L}_{A \rightarrow T} + \mathcal{L}_{T \rightarrow A})$	From $\Delta G_{AT} = \Delta G_{TA}$
Triplet fails	no pool normalization	Violates Axiom 2
BCE fails	independent pairs	Violates Axiom 5 (IIA)
Margin fails	hard gap	Violates Axiom 1 (soft selection)
Immune mapping	AIRE = hard negatives	Thymus runs contrastive learning

8. Checkpoint Questions

Q1. For a batch of $N = 8$ pairs, what is the untrained InfoNCE loss? What does this tell you about reading training curves?

Answer: The untrained loss is $\log N = \log 8 = 2.08$. At initialization the model is at chance, assigning $P = 1/8$ to the true pair, giving $\mathcal{L} = -\log(1/8) = \log 8 = 2.08$. This is a useful diagnostic: when you start training an SFM, the initial loss should be approximately $\log(\text{batch size})$. If your starting loss is far from $\log N$, something is wrong with initialization or the score scale. As training proceeds, the loss should fall below $\log N$ toward zero; how fast it falls and how low it goes is the physics signature discussed in later lectures.

Q2. In the trained example (§3), the diagonal score was 4.64 and off-diagonals were 0, giving $P = 0.981$. What diagonal score would give $P = 0.999$ (near-perfect retrieval) on the same 3-way batch?

Answer: We need $\exp(s)/(\exp(s) + 2) = 0.999$. Solving: $\exp(s) = 0.999(\exp(s) + 2) \Rightarrow \exp(s)(1 - 0.999) = 1.998 \Rightarrow \exp(s) = 1998 \Rightarrow s = \ln(1998) = 7.60$. So a diagonal score of about 7.6 gives 99.9% retrieval. The loss would be $-\log(0.999) = 0.001$. Note the diagonal score only had to roughly increase from 4.64 to 7.60 (about $1.6\times$) to move from 98.1% to 99.9% — the exponential makes the last bit of confidence cheap in score units but the loss is already very small, so there is little gradient signal left. This is part of why trained contrastive models plateau.

Q3. Explain precisely why binary cross-entropy violates IIA, using the structure of the loss.

Answer: BCE treats each agent–target pair as an independent binary classification: $P(\text{bind}_{ij}) = \sigma(s_{ij})$, with a separate sigmoid per pair and no shared denominator across the pool. The predicted probability for pair (i, j) depends only on s_{ij} and not on the scores s_{il} for other targets l . Consequently the relative odds between two targets, $P(\text{bind}_{ij})/P(\text{bind}_{ik})$, is $\sigma(s_{ij})/\sigma(s_{ik})$ — and as shown in WB1’s stress-test, this sigmoid ratio depends on the absolute scores, not just their difference, and there is no partition function coupling the targets into a competing pool. There is no sense in which adding or removing a target renormalizes the others. This is exactly the failure of IIA: BCE models the targets as non-competing independent events, whereas equilibrium binding is competitive selection from a shared pool. InfoNCE’s shared denominator is what enforces the competition; BCE lacks it.

Q4. The immune mapping (§6) identifies AIRE-driven negative selection with hard-negative mining. What practical lesson does this suggest for a student building a TCR-pMHC SFM?

Answer: The lesson is that hard-negative design is the decisive modeling choice, and biology tells you what the hard negatives are. For a TCR-pMHC SFM, the hard negatives should not be random non-binding peptides (those are easy negatives the model learns trivially); they should be peptides that are structurally or sequence-similar to true targets but do not bind — the computational analog of AIRE’s tissue-restricted self-antigens. Concretely: same-allele different-peptide pairs, peptides differing from a true binder by one or two anchor-position substitutions, or peptides from the same source protein that are not presented. Just as AIRE forces T cells to learn the hard self/non-self distinctions, well-chosen hard negatives force the SFM to learn the fine specificity boundaries that matter. This is why Lecture 9 treats hard-negative design as one of the five domain-expert decisions, and why getting it right matters more than any architectural tweak.

9. Connection to Future Lectures

- **Lecture 7:** The three levels of inevitability — unique equation (WB1), prescribed architecture (WB2), prescribed objective (WB3) — are consolidated into the five architecture conditions and the “prescribed vs. domain-specific” table. The symmetric InfoNCE loss derived here is one of the prescribed components. Level 3 alignment is defined.
- **Lecture 8:** The immune mapping previews the broader claim that molecular recognition systems across ten domains all implement the convergence equation. Thymic selection is the adaptive-immunity instance; the same structure recurs in every domain.
- **Lecture 9 (build):** Hard-negative design (§6, §8 Q4) is one of the five domain-expert decisions students make when building their own SFM. The untrained-loss diagnostic ($\mathcal{L} \approx \log N$, §3, §8 Q1) is a practical check students will use when they start their training runs.

- **Lecture 10 (evaluate):** The trained-vs-untrained loss trajectory ($1.10 \rightarrow 0.019$) is the basic shape of a contrastive training curve. The fine structure of that curve — how fast it falls, the train/test gap, OOD behavior — is the empirical physics signature students will observe in their own results and discuss in the workshop.